



# Chemistry

## 2022 TRIAL HSC EXAMINATION

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**General  
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A formulae sheet, data sheet and Periodic Table are provided at the back of this paper
- For questions in Section II, show all relevant working in questions involving calculations

**Total marks: 100****Section I – 20 marks (pages 3–10)**

- Attempt Questions 1–20
- Allow about 35 minutes for this section

**Section II – 80 marks in Booklet A and B (pages 11–30)**

- Attempt Questions 21– 32
- Allow about 2 hours and 25 minutes for this section in Booklet A and B

**Section I**  
**20 marks**

**Attempt Questions 1-20**  
**Allow about 35 minutes for this part**

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample:  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9

A  B  C  D 

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A  B  C  D 

correct

## Section I

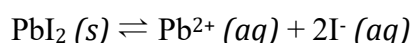
20 marks

Attempt Questions 1–20

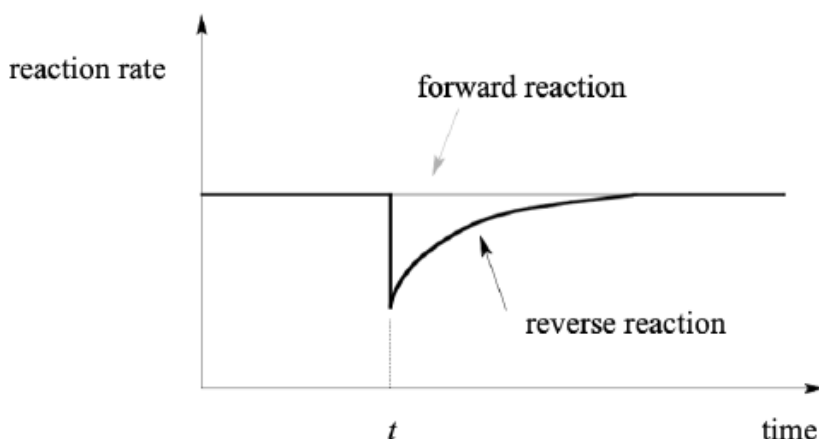
Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1 A small amount of solid lead iodide was added to a beaker of water, which was stirred. Most of the solid settled on the bottom of the beaker, but a little dissolved, establishing the equilibrium



The rates of the forward and reverse reactions were monitored over time, producing the graph shown below:



What happened at time  $t$ ?

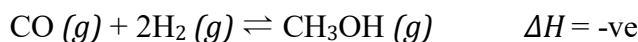
- (A) A small amount of solid  $\text{Pb}(\text{NO}_3)_2$  was added to the beaker.  
(B) A small amount of solid  $\text{KI}$  was added to the beaker.  
(C) A small amount of solid  $\text{PbI}_2$  was removed from the beaker.  
(D) A small amount of water was added to the beaker.
- 2 What is the  $\text{pH}$  of a  $0.010 \text{ mol L}^{-1}$  solution of a weak monoprotic acid that is 4.0% ionised?
- (A) 2.00  
(B) 2.40  
(C) 2.80  
(D) 3.40

- 3 Sulfuric acid ( $\text{H}_2\text{SO}_4$ ) and nitric acid ( $\text{HNO}_3$ ) are both strong acids. Ethanoic acid ( $\text{CH}_3\text{COOH}$ ) is a weak acid.

20.00 mL solutions of 0.10 mol/L concentration of each of these three acids were separately titrated with a 0.10 mol/L solution of sodium hydroxide ( $\text{NaOH}$ ).

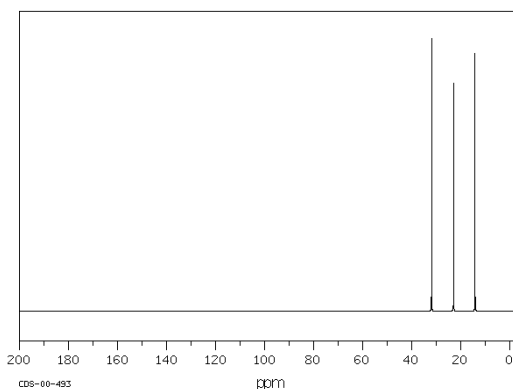
In order to react completely

- (A) all three acids would require the same amount of  $\text{NaOH}$ .
  - (B)  $\text{HNO}_3$  would require more  $\text{NaOH}$  than  $\text{CH}_3\text{COOH}$  but less than  $\text{H}_2\text{SO}_4$ .
  - (C)  $\text{H}_2\text{SO}_4$  and  $\text{HNO}_3$  would require the same amount of  $\text{NaOH}$  but  $\text{CH}_3\text{COOH}$  would require less.
  - (D)  $\text{CH}_3\text{COOH}$  and  $\text{HNO}_3$  would require the same amount of  $\text{NaOH}$  but  $\text{H}_2\text{SO}_4$  would require more.
- 4 It is known that carbon monoxide reacts exothermically with hydrogen gas to form methanol at  $400^\circ\text{C}$ , in the presence of a catalyst.



A mixture of carbon monoxide, hydrogen gas and methanol placed under conditions described above achieves equilibrium in a closed container. If the reaction temperature is changed to  $450^\circ\text{C}$ , which of the following statements is correct?

- (A) The total number of molecules in the container decreases.
  - (B) The reaction rates of both the forward and reverse reactions remain constant.
  - (C) The pressure of the gaseous mixture increases.
  - (D) Rate of formation of hydrogen decreases while the rate of decomposition of methanol increases.
- 5 Identify which one of the compounds given below gives the following  $^{13}\text{C}$  NMR spectrum:

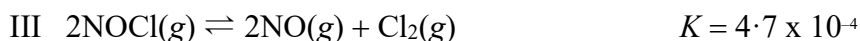
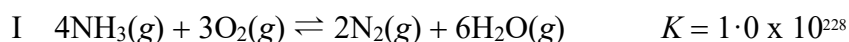


- (A) Prop-1-ene
- (B) Hexane
- (C) Butanoic acid
- (D) Ethanol

- 6 A student prepares a standard solution in the following way.
- 2.5 g of pure sodium hydroxide was dissolved into a clear, dry beaker using distilled water
  - The solution is funnelled into a volumetric flask, and the beaker and funnel set to one side without washing.
  - The volumetric flask was filled with distilled water, however the student went slightly over the calibration line.
  - The volumetric flask was shaken to ensure consistency

Despite the errors in the method, the student calculates the concentration of the standard solution. The standard solution is likely to be:

- (A) Weaker than calculated.  
(B) Stronger than calculated.  
(C) Less concentrated than calculated.  
(D) More concentrated than calculated.
- 7 Arrange the following reactions in order of their increasing tendency to reach completion.



(NOTE: For each reaction, the equilibrium constant was determined under different conditions.)

- (A) I, III, II, IV  
(B) III, II, IV, I  
(C) IV, II, III, I  
(D) I, IV, II, III
- 8 When acidic potassium permanganate is used to oxidise an alkene, a colour change occurs.

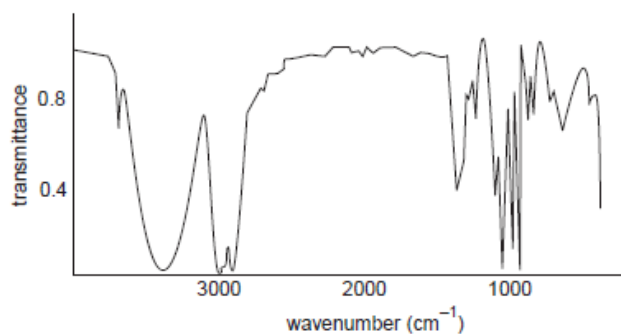
What colour change would be observed?

- (A) Colourless to purple  
(B) Purple to brown  
(C) Brown to colourless  
(D) Purple to colourless

- 9 2-Butanol is heated under reflux with acidified potassium dichromate solution. What is the major organic product?

(A) Butanal  
(B) Butyl butanoate  
(C) Butanone  
(D) Butanoic acid

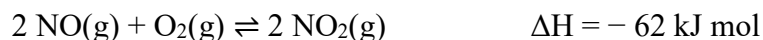
- 10 The infrared spectrum of an organic compound is shown below.



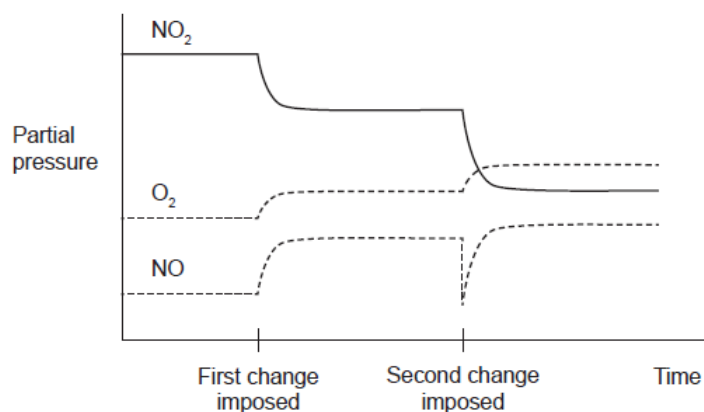
The compound could be

- (A) Propene  
(B) 2-Propanol  
(C) Propanoic acid  
(C) Propanone
- 11 When 1.27 g samples of the following substances are all treated with excess dilute hydrochloric acid, all give off carbon dioxide. Which gives off the greatest mass of carbon dioxide?
- (A) Lithium carbonate  
(B) Beryllium carbonate  
(C) Sodium carbonate  
(D) Magnesium carbonate

- 12 Nitrogen dioxide,  $\text{NO}_2(\text{g})$ , is formed when nitrogen monoxide,  $\text{NO}(\text{g})$ , undergoes oxidation as shown below.



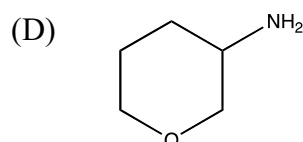
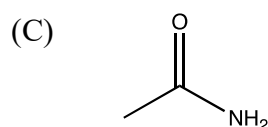
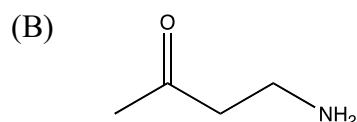
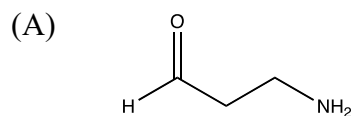
A change was imposed on an equilibrium gas mixture of  $\text{NO}_2$ ,  $\text{NO}$  and  $\text{O}_2$ . The mixture returned to equilibrium and another change was imposed. The following graph shows the effects of the two changes.



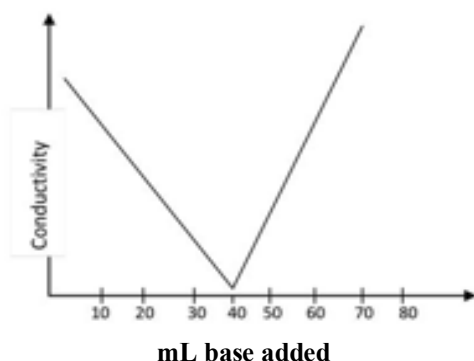
Identify the imposed changes that **best** account for the shape of the graph.

|     | First change                 | Second change                                     |
|-----|------------------------------|---------------------------------------------------|
| (A) | the temperature is decreased | the partial pressure of $\text{O}_2$ is increased |
| (B) | the temperature is decreased | the partial pressure of $\text{NO}$ is decreased  |
| (C) | the temperature is increased | the partial pressure of $\text{O}_2$ is increased |
| (D) | the temperature is increased | the partial pressure of $\text{NO}$ is decreased  |

- 13 Which one of the following compounds is both an amine and a ketone?



- 14 A conductometric titration was carried out using an acid and a base of similar concentration and the graph below was recorded.



The acid and base used could have been

- (A)  $\text{H}_2\text{SO}_4$  and  $\text{Ba}(\text{OH})_2$
  - (B)  $\text{H}_2\text{SO}_4$  and  $\text{NH}_3$
  - (C)  $\text{CH}_3\text{COOH}$  and  $\text{NH}_3$
  - (D)  $\text{CH}_3\text{COOH}$  and  $\text{Ba}(\text{OH})_2$
- 15 A colourless organic liquid compound is tested for the presence of functional groups. 2 mLs of the liquid are poured into three separate tests tubes. A reagent is added to each test tube then shaken.

| <i>Test</i> | <i>Reagent Added</i>                      | <i>Observation made after shaking</i>                |
|-------------|-------------------------------------------|------------------------------------------------------|
| 1           | 2 drops of Bromine water                  | Yellowish-brown coloured solution                    |
| 2           | 3 drops of acidified potassium dichromate | Yellowish-orange coloured solution                   |
| 3           | 1 gram of sodium hydrogen carbonate       | Colourless solution, small bubbles of colourless gas |

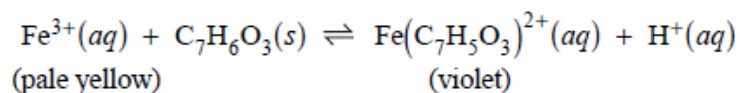
What functional group(s) may be present in this compound?

- (A) Double bonds
- (B) Carboxylic acid
- (C) Double bonds and hydroxyl group
- (D) Hydroxyl group bonded to a secondary carbon

**16** Which row of the table below correctly matches the reaction type with its correct reactants, catalyst and products?

|     | Reaction Type | Reactants                                                             | Catalyst                             | Products                                       |
|-----|---------------|-----------------------------------------------------------------------|--------------------------------------|------------------------------------------------|
| (A) | Hydration     | $\text{H}_2\text{C} = \text{CH}_2 + \text{H}_2\text{O}$               | $\text{H}_2\text{SO}_4$              | $\text{HOCH}_2\text{CH}_2\text{OSO}_3\text{H}$ |
| (B) | Hydration     | $\text{H}_2\text{C} = \text{CH}_2$                                    | concentrated $\text{H}_2\text{SO}_4$ | $\text{CH}_3\text{CH}_2\text{OH}$              |
| (C) | Addition      | $\text{H}_2\text{C} = \text{CH}_2 + \text{H}_2\text{O} + \text{Br}_2$ | $\text{H}_2\text{SO}_4$              | $\text{BrCH}_2\text{CH}_2\text{Br}$            |
| (D) | Addition      | $\text{H}_2\text{C} = \text{CH}_2 + \text{H}_2$                       | Nickel                               | $\text{CH}_3\text{CH}_3$                       |

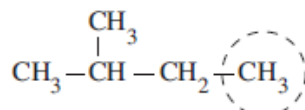
**17** Iron(III) forms a variety of coloured complex ions. One such example is that produced by the reaction with salicylic acid ( $\text{C}_7\text{H}_6\text{O}_3$ ):



The yield of the violet complex,  $\text{Fe}(\text{C}_7\text{H}_5\text{O}_3)^{2+}$ , may be decreased by

- (A) adding  $\text{C}_7\text{H}_6\text{O}_3(s)$  to the solution.  
 (B) adding  $\text{Fe}^{3+}$  ions to the solution.  
 (C) lowering the pH of the solution.  
 (D) raising the pH of the solution.

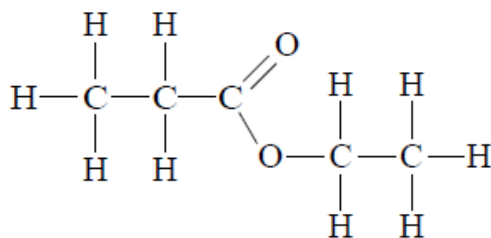
**18** The structure of 2-methylbutane is shown. A CH<sub>3</sub> group is circled



Which of the following splitting patterns would be observed in the  $^1\text{H}$  NMR spectrum for the  $\text{CH}_3$  group circled?

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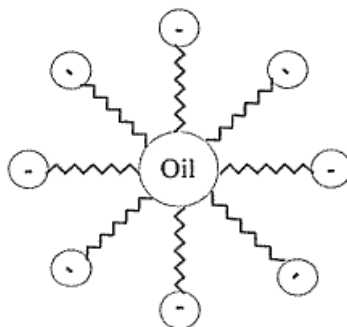
19



The species that produces the molecular ion peak in the mass spectrum of this compound is

- (A)  $[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3]^+$
- (B)  $[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3]^{2+}$
- (C)  $[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3]^-$
- (D)  $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$

20 Which statement correctly describes the diagram below?



- (A) The hydrophilic head of the soap molecule is polar and is attracted to water molecules. The hydrophobic tail is non-polar and forms micelles trapping the oil within the micelle.
- (B) The hydrophilic head of the soap molecule is non-polar and is attracted to water molecules. The hydrophobic tail is polar and forms micelles trapping the oil within the micelle.
- (C) The hydrophilic head of the soap molecule is polar and is attracted to oil molecules. The hydrophobic tail is non-polar and forms micelles trapping the water within the micelle.
- (D) The hydrophobic head of the soap molecule is non-polar and is attracted to water molecules. The hydrophilic tail is non-polar and forms micelles trapping the oil within the micelle.

# Chemistry

## 2022 TRIAL EXAMINATION

### Section II

80 marks

Attempt Questions 21- 32

Allow about 2 hour and 25 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Section II

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#### Question 21 (3 Marks)

Calculate the minimum concentration of silver nitrate that will cause precipitation of silver bromide when 25.0 mL of the silver nitrate solution is added to 25.0 mL of 0.020 mol/L<sup>-1</sup> sodium bromide.

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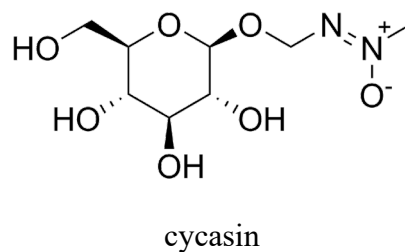
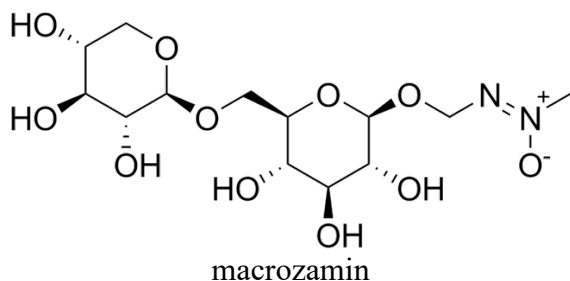
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**Question 22 (3 marks)**

The seeds of cycad plants are used by Aboriginal and Torres Strait Islander peoples to make bread. However, these seeds contain toxins, TWO of which are illustrated below.



Explain the process used to remove these toxins with reference to the structure of each molecule.

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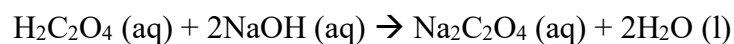
**Question 23 (7 marks)**

High strength aspirin ( $\text{C}_9\text{H}_8\text{O}_4$ ) tablets were purchased overseas by the schools chemistry teacher. Puzzled by the low cost, he encouraged their students to perform a chemical analysis on the concentration of Aspirin in each tablet.

Students were given 6 tablets each and were instructed to use sodium hydroxide solution with an approximate concentration of 0.100 mol/L, which was standardised using oxalic acid ( $\text{COOH}$ )<sub>2</sub>.

Four flasks were prepared each containing a mixture of 25.00 mL of water and 10.00 mL of ethanol. 2 aspirin tablets were dissolved in each flask for titration against the sodium hydroxide.

- (a) Three 25.00 mL samples of a 0.1034 mol/L solution of oxalic acid were titrated to standardise the NaOH solution according to the following equation:



The average volume required for neutralisation was 25.75 mL. Calculate the molarity of the NaOH solution.

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- (b) What is the endpoint of a titration?

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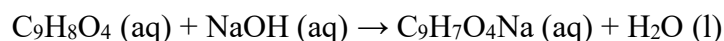
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**Question 23 continues on the next page**

**Question 23 continued**

- (b) Four flasks were prepared each containing a mixture of 25.00 mL of water and 10.00 mL of ethanol. Two aspirin tablets was dissolved in each flask. The aspirin in each solution was titrated with the standardised NaOH solution according to the following equation:



The students achieved the following titration results:

| Trial | Volume (mL) |
|-------|-------------|
| 1     | 16.55       |
| 2     | 16.50       |
| 3     | 16.60       |
| 4     | 16.85       |

The packaging claims the tablets contain 150.0 mg of aspirin per tablet.

Using the information provided, calculate the mass (mg) of aspirin per tablet, and justify whether the claim is true.

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**Question 24 (4 marks)**

- (a) What is the maximum pH which can be obtained by dissolving  $\text{Mg}(\text{OH})_2$  in pure water? 2

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- (b) How does this compare with the maximum pH which can be obtained with NaOH? 2  
(The solubility of NaOH is 1.20 g/mL at 25°C)

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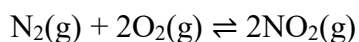
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**Question 25 (14 marks)**

Nitrogen dioxide is toxic to humans when inhaled and is a significant component of air pollution.

It can be formed by the combustion of nitrogen in the air at high temperatures; firstly forming nitric oxide NO(g) and on further oxidation, forming nitrogen dioxide, NO<sub>2</sub>(g).

The overall equation for this process is given here:

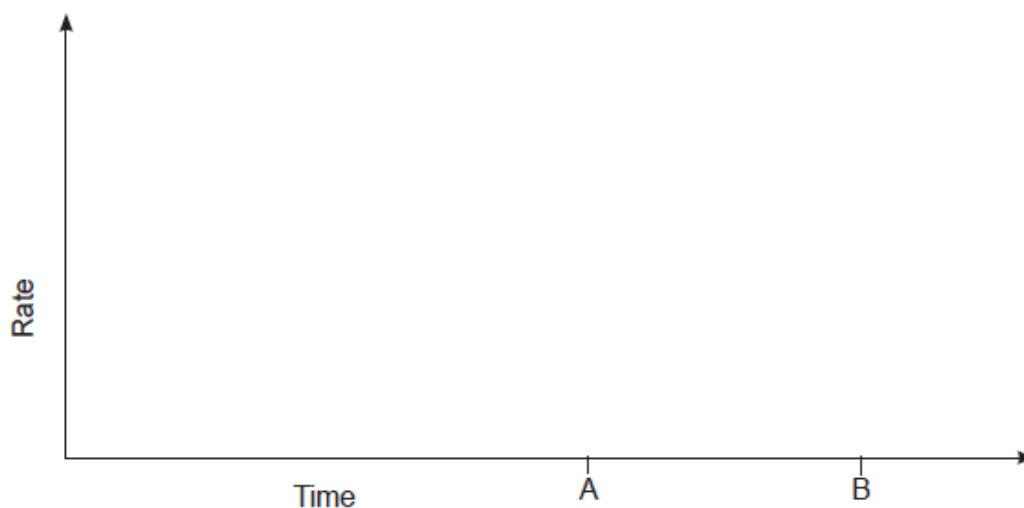


The following questions relate to the equilibrium system represented by this equation.

- (a) Write the equilibrium expression for this reaction when it is in equilibrium. 1

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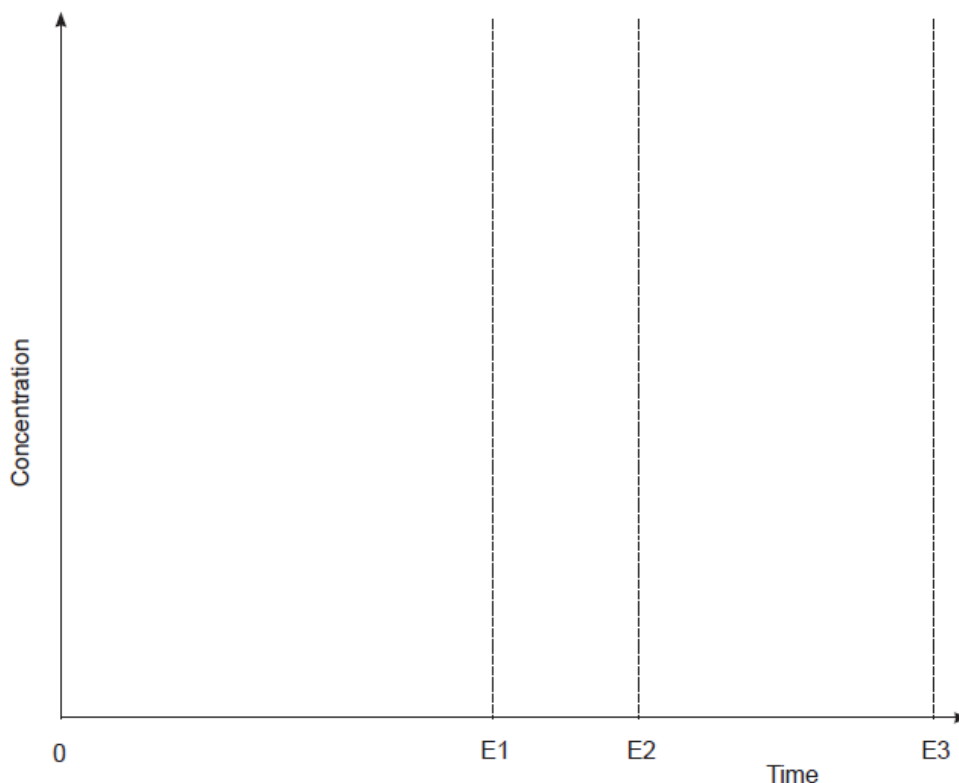
- (b) On the axes below, draw the forward (—) and reverse (- - -) reaction rates, 3  
(i) starting at the moment the oxygen and nitrogen gases begin to react with each other until after equilibrium has been established at time A. Continue the graph until time B.



- (ii) On the same axes above, draw and label clearly the effect of conducting the same 2  
reaction at a higher temperature.

- (c) On the axes below, draw separate curves to show how the concentrations of the three gases change with time, starting at the moment the oxygen and nitrogen gases begin to react with each other until the system reaches equilibrium at Time E1. Continue the graph from Time E1 to Time E2. Assume that the initial concentrations of oxygen and nitrogen are identical. 5

Label clearly the line for each gas.



- (d) At Time E2 shown on the axis, the reaction vessel is doubled in volume, and the system is then again allowed to reach equilibrium at Time E3. 3

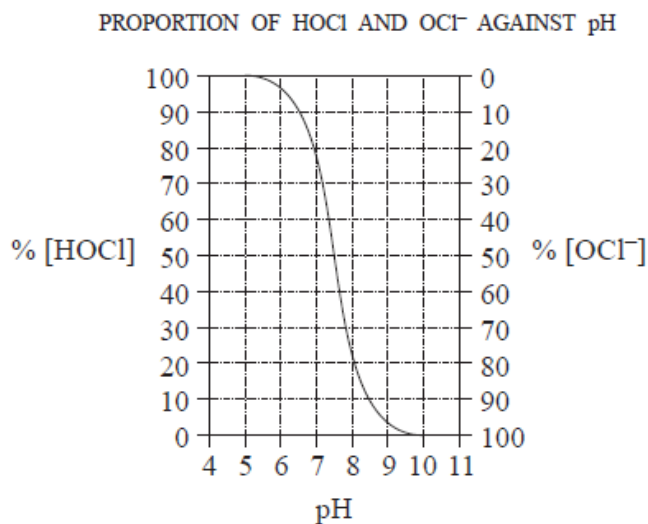
On the same graph above, show how the concentrations of the three gases would change in response to the change in volume, from Time E2 until equilibrium is re-established at Time E3.

**Question 26 (5 marks)**

Hypochlorous acid (HOCl) is often used in swimming pools as a means of killing bacteria. The HOCl dissociates as shown:



The curve below shows the distribution of [HOCl] and [OCl<sup>-</sup>] in water at 25°C.



- (a) What is the effect on pH when OCl<sup>-</sup> ions are added to the water? 1

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- (b) At pH = 8.5, what is the ratio of [OCl<sup>-</sup>] : [HOCl]? 1

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- (c) Using your answer in part (b), calculate  $K_a$  for hypochlorous acid. 3

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**Question 27 (5 marks)**

Two neighbouring factories produce large quantities of waste that they need to dispose of safely. One waste is acidic and the other is basic.

25.0 mL of waste A required 29.5 mL of 0.233 mol/L sulfuric acid for exact neutralisation.

25.0 mL of waste B required 32.8 mL of 0.274 mol/L sodium hydroxide for exact neutralisation.

What volume of waste A should be mixed with 100.0 L of waste B in order to produce neutral waste? **5**

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## Section II extra writing space.

If you use this space indicate clearly which question you are answering.

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NESA number

**SYDNEY BOYS HIGH SCHOOL**  
**CHEMISTRY**  
**2022 TRIAL HSC EXAMINATION**

**Section II - Booklet B**  
**Answer questions 30-35**

Booklet B Total:

**Question 28 (4 marks)**

A student has three unlabelled test tubes. They contain three different but apparently identical colourless liquids. One is 1-butanol, another is 2-methyl-2-propanol, and the third is 1-hexene.

Describe the chemical tests the student should carry out to determine which liquid is in each test tube. **4**

In your answer, include expected observations and relevant equations.

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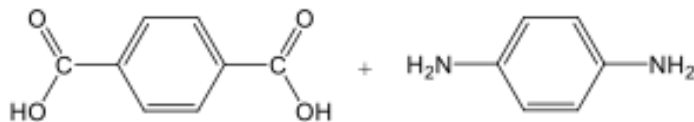
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### Question 29 (4 marks)

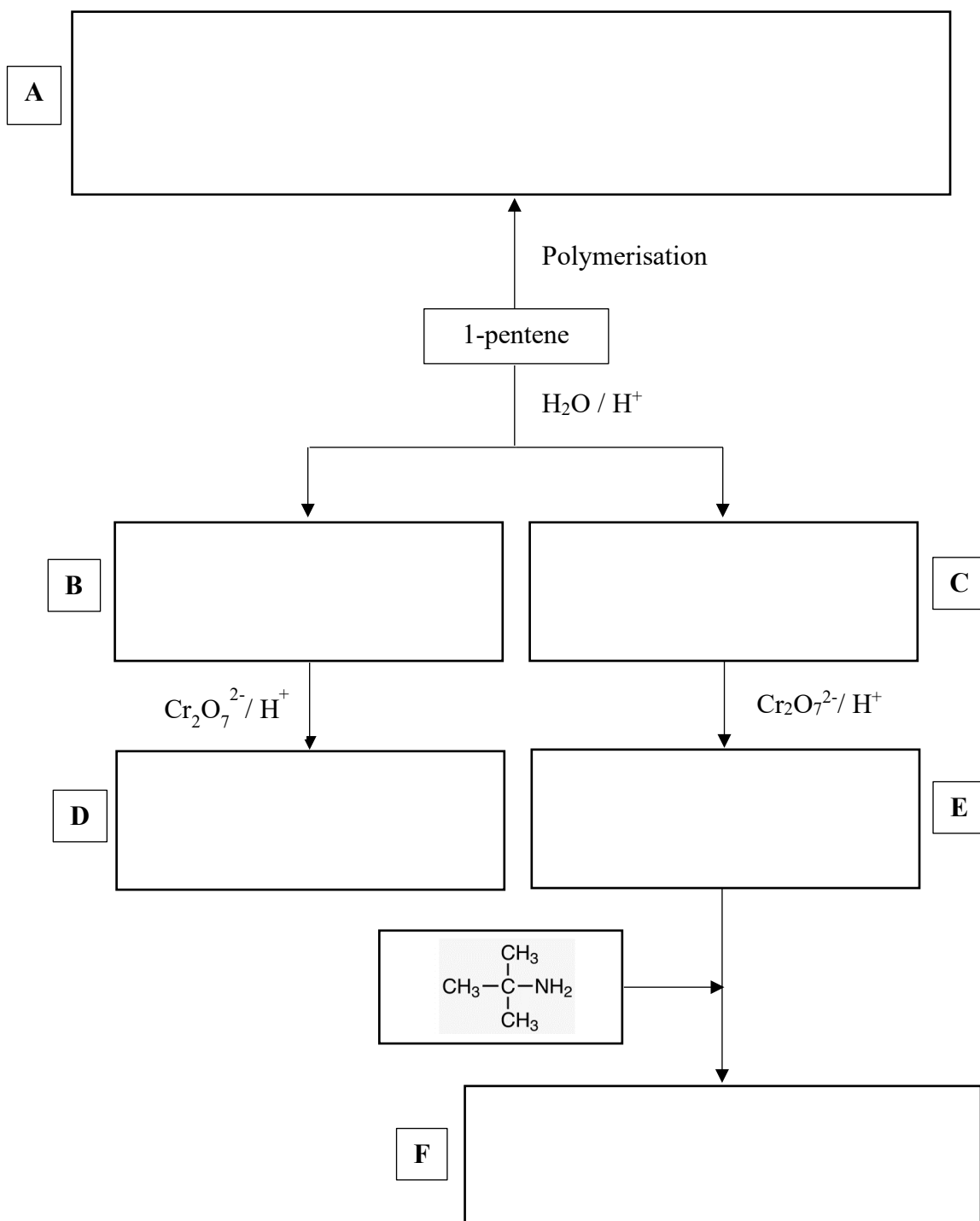
Kevlar is a strong nylon polymer prepared by the reaction shown below. Draw a segment of Kevlar and explain the strength and subsequent uses of the polymer based on its structure.

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[illegible]

**Question 30 (15 marks)**

A student designed and carried out the following synthetic pathway during his studies of chemical reactions on organic compounds. His chosen starting material was 1-pentene.



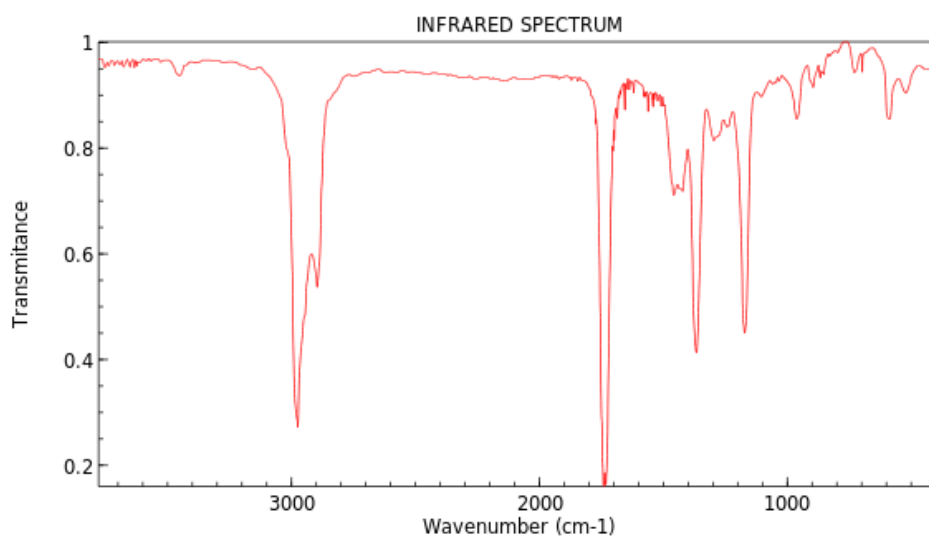
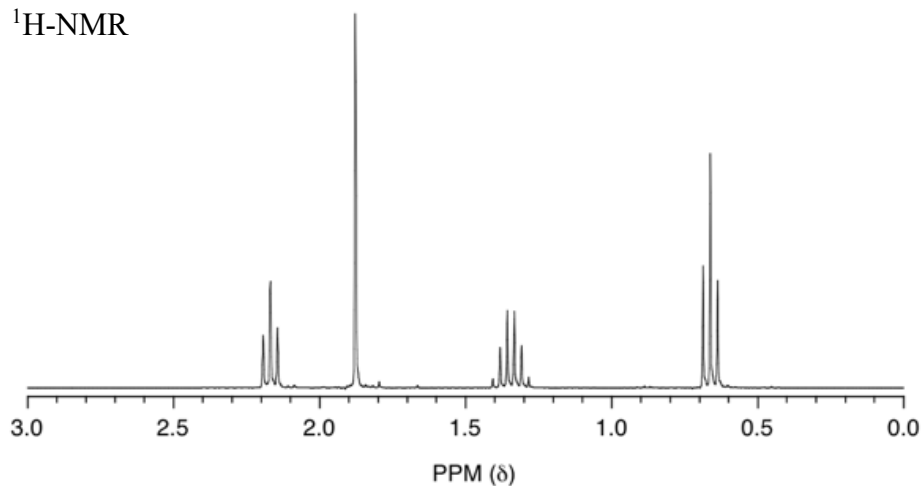
- (a) Draw the structural formulas of compounds B to F in the boxes provided. **5**
- (b) Draw the semi-structural formula, containing 3 monomers, of compound A in the box provided. **1**

**Question 30 continues on the following page.**

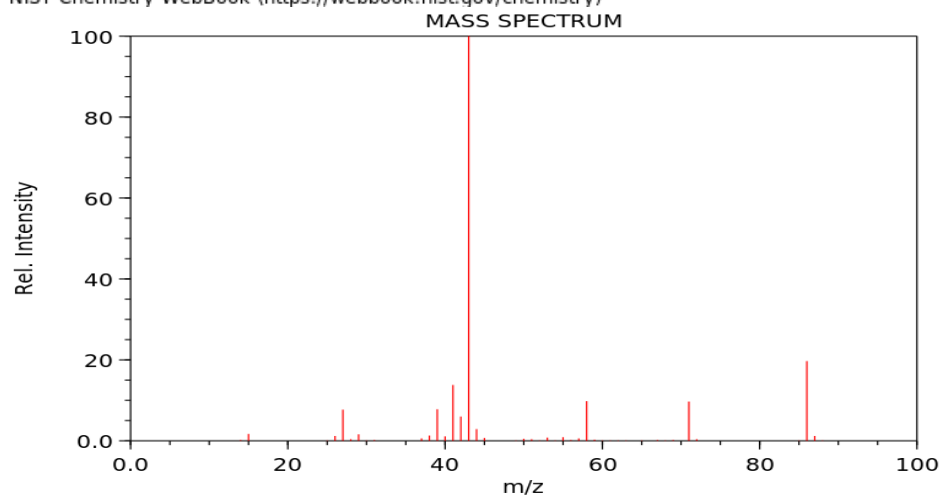
**Question 30 continues.**

- (c) To distinguish between compounds D and E, the student used spectroscopic analysis to determine the structure of the compound produced. He was provided with the following spectra.

$^1\text{H}$ -NMR



NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)



NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)

**Question 30 continues on the following page.**

**Question 30 continues.**

Identify which compound from the previous synthetic pathway corresponds to the provided spectra. Justify your choice using the information from each spectrum. **6**

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**Question 30 continued on the following page.**

Question 30 continued.

<sup>1</sup>H NMR chemical shift data

| Type of proton                                           | δ/ppm    |
|----------------------------------------------------------|----------|
| Si(CH <sub>3</sub> ) <sub>4</sub> (TMS)                  | 0        |
| R—CH <sub>3</sub>                                        | 0.7–1.3  |
| R—CH <sub>2</sub> —R                                     | 1.2–1.5  |
| R—CHR <sub>2</sub>                                       | 1.5–2.0  |
| H <sub>3</sub> C—CO— (aldehydes, ketones or esters)      | 2.0–2.5  |
| —CH—CO— (aldehydes, ketones or esters)                   | 2.1–2.6  |
| H <sub>3</sub> C—O— (alcohols or esters)                 | 3.2–4.0  |
| —CH—O— (alcohols or esters)                              | 3.3–5.1  |
| R <sub>2</sub> —CH <sub>2</sub> —O— (alcohols or esters) | 3.5–5.0  |
| R—OH                                                     | 1–6      |
| R <sub>2</sub> C=CHR (alkene)                            | 4.5–7.0  |
| R—CHO (aldehyde)                                         | 9.4–10.0 |
| R—COOH                                                   | 9.0–13.0 |

1

- (d) The IUPAC name for Compound D is:

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- (e) In a <sup>13</sup>C NMR spectrum of Compound D, how many signals would you expect? Why?

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**Question 31 (5 marks)**

Some properties of but-1-ene, 1-fluoropropane, propan-1-ol and ethanamide are shown in the table.

| <i>Compound</i> | <i>Molar mass (<math>\text{g mol}^{-1}</math>)</i> | <i>Solubility in water</i> |
|-----------------|----------------------------------------------------|----------------------------|
| but-1-ene       | 56                                                 | insoluble                  |
| 1-fluoropropane | 62                                                 | slightly soluble           |
| propan-1-ol     | 60                                                 | soluble                    |
| ethanamide      | 59                                                 | soluble                    |

Explain the different water solubilities of these compounds. Support your answer with at least ONE labelled diagram.

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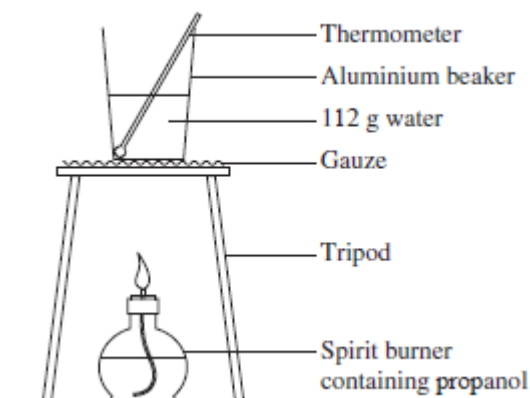
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**Question 32 (11 marks)**

The following apparatus was used in an experiment to determine the molar enthalpy of combustion of propanol.



- (a) Calculate the experimental molar enthalpy of combustion ( $\Delta H_c$ ) of propanol when 0.543 g propanol was used to raise the water temperature from 19.0°C to 37.5°C. 4  
 $MM(C_3H_7OH) = 60.10 \text{ g mol}^{-1}$

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- (b) Upon replication, the molar enthalpy of combustion obtained in the experiment was consistently much lower than the accepted value. 2

Explain ONE change that could be made to the experiment that would improve the accuracy of the obtained value.

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**Question 32 continues on the following page.**

**Question 32 continues.**

The molar heat of combustion of propan-1-ol,  $\text{C}_3\text{H}_7\text{OH}$ , is  $2021 \text{ kJ mol}^{-1}$ .

- (c) Calculate the mass of  $\text{C}_3\text{H}_7\text{OH}$ , in tonnes, required to produce 9 605 MJ of energy. **3**

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One way of minimising the greenhouse effect is to minimise the amount of carbon dioxide formed per kilojoule of energy produced. **2**

- (d) Using the theoretical  $\Delta H_c$ , calculate the amount of energy released, in kJ, per mole of carbon dioxide produced during the complete combustion of propan-1-ol.

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**End of paper**

## Section II extra writing space.

If you use this space indicate clearly which question you are answering.

[illegible]

## Section II extra writing space.

If you use this space indicate clearly which question you are answering.

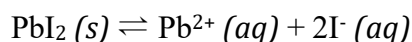
This image shows a full page of white paper with horizontal dotted lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.



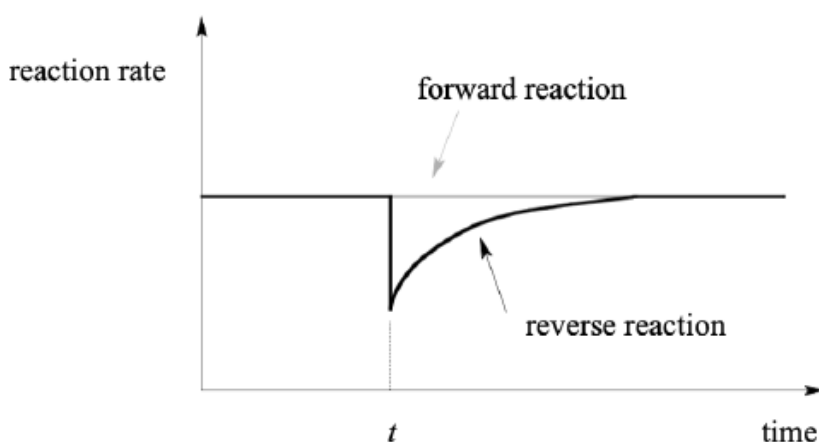
# SBHS TRIAL 2022 – MARKING CRITERIA

Use the multiple-choice answer sheet for Questions 1–20.

- 1 A small amount of solid lead iodide was added to a beaker of water, which was stirred. Most of the solid settled on the bottom of the beaker, but a little dissolved, establishing the equilibrium



The rates of the forward and reverse reactions were monitored over time, producing the graph shown below:



What happened at time  $t$ ?

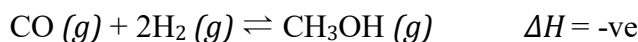
- (A) A small amount of solid  $\text{Pb}(\text{NO}_3)_2$  was added to the beaker.  
(B) A small amount of solid  $\text{KI}$  was added to the beaker.  
(C) A small amount of solid  $\text{PbI}_2$  was removed from the beaker.  
(D) A small amount of water was added to the beaker.
- 2 What is the  $\text{pH}$  of a  $0.010 \text{ mol L}^{-1}$  solution of a weak monoprotic acid that is 4.0% ionised?
- (A) 2.00  
(B) 2.40  
(C) 2.80  
(D) 3.40

- 3 Sulfuric acid ( $\text{H}_2\text{SO}_4$ ) and nitric acid ( $\text{HNO}_3$ ) are both strong acids. Ethanoic acid ( $\text{CH}_3\text{COOH}$ ) is a weak acid.

20.00 mL solutions of 0.10 mol/L concentration of each of these three acids were separately titrated with a 0.10 mol/L solution of sodium hydroxide ( $\text{NaOH}$ ).

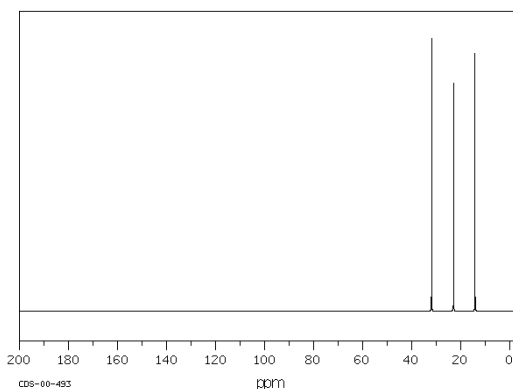
In order to react completely

- (A) all three acids would require the same amount of  $\text{NaOH}$ .  
(B)  $\text{HNO}_3$  would require more  $\text{NaOH}$  than  $\text{CH}_3\text{COOH}$  but less than  $\text{H}_2\text{SO}_4$ .  
(C)  $\text{H}_2\text{SO}_4$  and  $\text{HNO}_3$  would require the same amount of  $\text{NaOH}$  but  $\text{CH}_3\text{COOH}$  would require less.  
(D)  $\text{CH}_3\text{COOH}$  and  $\text{HNO}_3$  would require the same amount of  $\text{NaOH}$  but  $\text{H}_2\text{SO}_4$  would require more.
- 4 It is known that carbon monoxide reacts exothermically with hydrogen gas to form methanol at  $400^\circ\text{C}$ , in the presence of a catalyst.



A mixture of carbon monoxide, hydrogen gas and methanol placed under conditions described above achieves equilibrium in a closed container. If the reaction temperature is changed to  $450^\circ\text{C}$ , which of the following statements is correct?

- (A) The total number of molecules in the container decreases.  
(B) The reaction rates of both the forward and reverse reactions remain constant.  
(C) The pressure of the gaseous mixture increases.  
(D) Rate of formation of hydrogen decreases while the rate of decomposition of methanol increases.
- 5 Identify which one of the compounds given below gives the following  $^{13}\text{C}$  NMR spectrum:



- (A) Prop-1-ene  
(B) Hexane  
(C) Butanoic acid  
(D) Ethanol

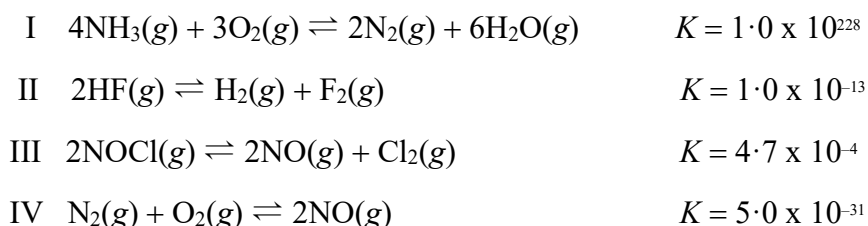
6 A student prepares a standard solution in the following way.

- i. 2.5 g of pure sodium hydroxide was dissolved into a clear, dry beaker using distilled water
- ii. The solution is funnelled into a volumetric flask, and the beaker and funnel set to one side without washing.
- iii. The volumetric flask was filled with distilled water, however the student went slightly over the calibration line.
- iv. The volumetric flask was shaken to ensure consistency

Despite the errors in the method, the student calculates the concentration of the standard solution. The standard solution is likely to be:

- (A) Weaker than calculated.  
(B) Stronger than calculated.  
(C) Less concentrated than calculated.  
(D) More concentrated than calculated.

7 Arrange the following reactions in order of their increasing tendency to reach completion.



(NOTE: For each reaction, the equilibrium constant was determined under different conditions.)

- (A) I, III, II, IV  
(B) III, II, IV, I  
(C) IV, II, III, I  
(D) I, IV, II, III

8 When acidic potassium permanganate is used to oxidise an alkene, a colour change occurs.

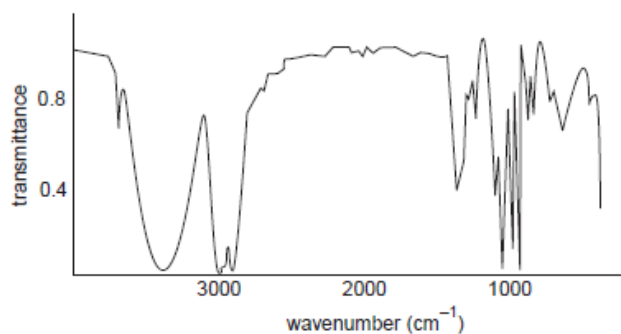
What colour change would be observed?

- (A) Colourless to purple  
(B) Purple to brown  
(C) Brown to colourless  
(D) Purple to colourless

- 9 2-Butanol is heated under reflux with acidified potassium dichromate solution. What is the major organic product?

(A) Butanal  
(B) Butyl butanoate  
(C) Butanone  
(D) Butanoic acid

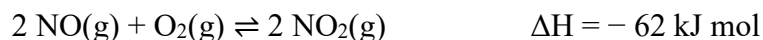
- 10 The infrared spectrum of an organic compound is shown below.



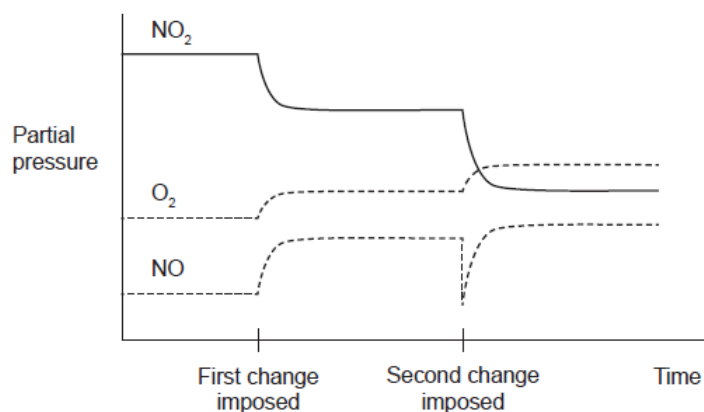
The compound could be

- (A) Propene  
(B) 2-Propanol  
(C) Propanoic acid  
(C) Propanone
- 11 When 1.27 g samples of the following substances are all treated with excess dilute hydrochloric acid, all give off carbon dioxide. Which gives off the greatest mass of carbon dioxide?
- (A) Lithium carbonate  
(B) Beryllium carbonate  
(C) Sodium carbonate  
(D) Magnesium carbonate

- 12 Nitrogen dioxide,  $\text{NO}_2(\text{g})$ , is formed when nitrogen monoxide,  $\text{NO}(\text{g})$ , undergoes oxidation as shown below.



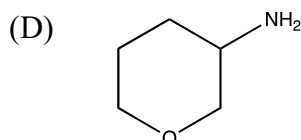
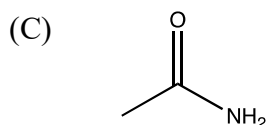
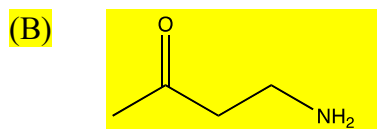
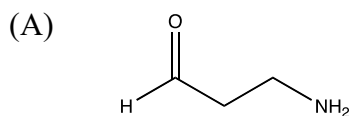
A change was imposed on an equilibrium gas mixture of  $\text{NO}_2$ ,  $\text{NO}$  and  $\text{O}_2$ . The mixture returned to equilibrium and another change was imposed. The following graph shows the effects of the two changes.



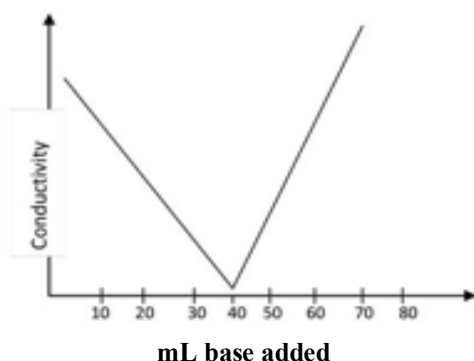
Identify the imposed changes that **best** account for the shape of the graph.

|     | First change                 | Second change                                     |
|-----|------------------------------|---------------------------------------------------|
| (A) | the temperature is decreased | the partial pressure of $\text{O}_2$ is increased |
| (B) | the temperature is decreased | the partial pressure of $\text{NO}$ is decreased  |
| (C) | the temperature is increased | the partial pressure of $\text{O}_2$ is increased |
| (D) | the temperature is increased | the partial pressure of $\text{NO}$ is decreased  |

- 13 Which one of the following compounds is both an amine and a ketone?



- 14 A conductometric titration was carried out using an acid and a base of similar concentration and the graph below was recorded.



The acid and base used could have been

- (A)  $\text{H}_2\text{SO}_4$  and  $\text{Ba}(\text{OH})_2$   
 (B)  $\text{H}_2\text{SO}_4$  and  $\text{NH}_3$   
 (C)  $\text{CH}_3\text{COOH}$  and  $\text{NH}_3$   
 (D)  $\text{CH}_3\text{COOH}$  and  $\text{Ba}(\text{OH})_2$
- 15 A colourless organic liquid compound is tested for the presence of functional groups. 2 mLs of the liquid are poured into three separate tests tubes. A reagent is added to each test tube then shaken.

| Test | Reagent Added                             | Observation made after shaking                       |
|------|-------------------------------------------|------------------------------------------------------|
| 1    | 2 drops of Bromine water                  | Yellowish-brown coloured solution                    |
| 2    | 3 drops of acidified potassium dichromate | Yellowish-orange coloured solution                   |
| 3    | 1 gram of sodium hydrogen carbonate       | Colourless solution, small bubbles of colourless gas |

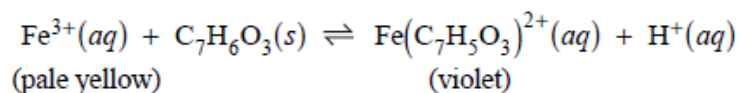
What functional group(s) may be present in this compound?

- (A) Double bonds  
 (B) Carboxylic acid  
 (C) Double bonds and hydroxyl group  
 (D) Hydroxyl group bonded to a secondary carbon

**16** Which row of the table below correctly matches the reaction type with its correct reactants, catalyst and products?

|     | Reaction Type | Reactants                                                             | Catalyst                             | Products                                       |
|-----|---------------|-----------------------------------------------------------------------|--------------------------------------|------------------------------------------------|
| (A) | Hydration     | $\text{H}_2\text{C} = \text{CH}_2 + \text{H}_2\text{O}$               | $\text{H}_2\text{SO}_4$              | $\text{HOCH}_2\text{CH}_2\text{OSO}_3\text{H}$ |
| (B) | Hydration     | $\text{H}_2\text{C} = \text{CH}_2$                                    | concentrated $\text{H}_2\text{SO}_4$ | $\text{CH}_3\text{CH}_2\text{OH}$              |
| (C) | Addition      | $\text{H}_2\text{C} = \text{CH}_2 + \text{H}_2\text{O} + \text{Br}_2$ | nil                                  | $\text{BrCH}_2\text{CH}_2\text{Br}$            |
| (D) | Addition      | $\text{H}_2\text{C} = \text{CH}_2 + \text{H}_2$                       | Nickel                               | $\text{CH}_3\text{CH}_3$                       |

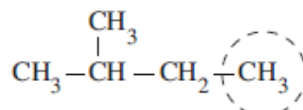
**17** Iron(III) forms a variety of coloured complex ions. One such example is that produced by the reaction with salicylic acid ( $\text{C}_7\text{H}_6\text{O}_3$ ):



The yield of the violet complex,  $\text{Fe}(\text{C}_7\text{H}_5\text{O}_3)^{2+}$ , may be decreased by

- (A) adding  $\text{C}_7\text{H}_6\text{O}_3(s)$  to the solution.  
(B) adding  $\text{Fe}^{3+}$  ions to the solution.  
(C) lowering the pH of the solution.  
(D) raising the pH of the solution.

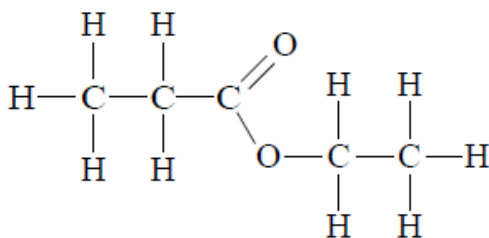
**18** The structure of 2-methylbutane is shown. A CH<sub>3</sub> group is circled



Which of the following splitting patterns would be observed in the  $^1\text{H}$  NMR spectrum for the  $\text{CH}_3$  group circled?

- (A) 
- (B) 
- (C) 
- (D) 

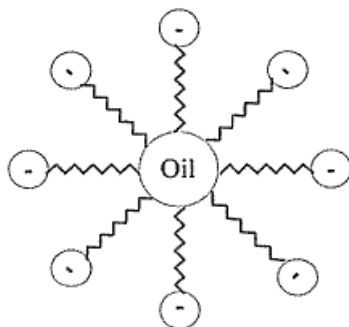
19



The species that produces the molecular ion peak in the mass spectrum of this compound is

- (A)  $[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3]^+$   
 (B)  $[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3]^{2+}$   
 (C)  $[\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3]^-$   
 (D)  $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$

**20** Which statement correctly describes the diagram below?



- (A) The hydrophilic head of the soap molecule is polar and is attracted to water molecules. The hydrophobic tail is non-polar and forms micelles trapping the oil within the micelle.
- (B) The hydrophilic head of the soap molecule is non-polar and is attracted to water molecules. The hydrophobic tail is polar and forms micelles trapping the oil within the micelle.
- (C) The hydrophilic head of the soap molecule is polar and is attracted to oil molecules. The hydrophobic tail is non-polar and forms micelles trapping the water within the micelle.
- (D) The hydrophobic head of the soap molecule is non-polar and is attracted to water molecules. The hydrophilic tail is non-polar and forms micelles trapping the oil within the micelle.

# Chemistry

## 2022 TRIAL EXAMINATION

### Section II

80 marks

Attempt Questions 21- 32

Allow about 2 hour and 25 minutes for this part

Answer the questions in the spaces provided.

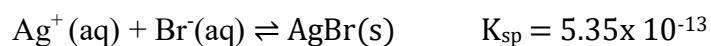
Show all relevant working in questions involving calculations.

Section II

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#### Question 21 (3 Marks)

3



$$[\text{Br}^-] = 0.02/2 = 0.01 \text{ mol/L} \quad (25\text{mL added to } 25\text{mL} \rightarrow \text{conc halved})$$

$$K_{\text{sp}} = [\text{Ag}^+][\text{Br}^-]$$

$$[\text{Ag}^+] = 5.35 \times 10^{-13} / 0.01 = 5.35 \times 10^{-11} \text{ mol/L (after dilution)}$$

$$\begin{aligned} \text{Therefore } [\text{AgNO}_3] \text{ required be added} &= 2 \times 5.35 \times 10^{-11} \\ &= 1.07 \times 10^{-10} \text{ mol/L} \end{aligned}$$

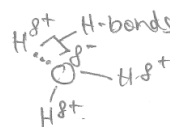
(2x as initial mixture which is more concentrated)

#### Question 22 (3 marks)

**Sample answer**

Aboriginal and Torres Strait Islander peoples ~~use~~ ground the cycad seeds into a fine powder firstly to increase surface area for leaching. The seeds would be soaked ~~in~~ in running water in rivers and streams in leaching bags. The toxins, macrozamin and cycasin will dissolve readily in water as they are highly soluble. This is because the toxins ~~are~~ contain ~~polar~~ <sup>multiple polar hydroxyl</sup> groups which due to the very electronegative oxygen atom can form strong hydrogen bonds with water,  $R-O^{\delta-}$  to make them dissolve in it. Running water constantly removes aqueous ~~cycas~~ toxins so shifts the dissolution ~~equilibrium~~ <sup>to the right</sup> constantly ensuring almost complete ~~diss~~ dissolution and removal of toxins from seeds, safe for ~~eat~~ as food.

6 OH groups  
for  
macrozamin,  
4 OH groups  
for cycasin



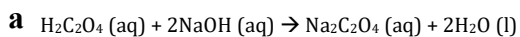
Leaching involves submerging the grounded up powder from the cycad seeds inside a woven bag in running water for up to 4 weeks, depending on the type of seed.

Running water is necessary so that the water soluble toxins are washed away and do not reach equilibrium. Each toxin molecule is polar due to the numerous OH functional groups and polar O atoms, hence making them soluble in water which is also polar by forming H-bonding with water molecules

Furthermore, grinding the seeds increases the surface area available for the water to pass through and remove more of the toxin. The remaining carbohydrate is insoluble and can be cooked to make bread.

**Question 23 (7 marks)**

( *Sample answer*



)

$$\begin{aligned} n(\text{H}_2\text{C}_2\text{O}_4) &= cv \\ &= 0.1034 \times 0.25 \\ &= 2.85 \times 10^{-3} \text{ mol} \\ &= n(\text{NaOH}) \times 2 \\ n(\text{NaOH}) &= 0.00517 \text{ mol (1:2 ratio)} \\ c(\text{NaOH}) &= \frac{c}{v} = \frac{0.00517}{0.02575} \\ &= 0.2008 \text{ mol L}^{-1} \end{aligned}$$

2

( What is the endpoint of a titration?

1

**b**

|   | <i>Criteria</i>    | <i>Marks</i> |
|---|--------------------|--------------|
| ) | Correct definition | 1            |

The point during a titration when the indicator first produces a **permanent** colour change and this indicates that equivalence point has been reached.

NB- With reference to colour change, it is not acceptable to state:

- 1) the first visible or just visible
- 2) observable
- 3) as soon as the indicator changes
- 4) slight

(

**c**

)

*Sample answer*

$$\begin{aligned} \text{Avg. vol.} &= 16.55 \text{ mL (trial 4 is an outlier)} \\ n(\text{NaOH}) &= cv \\ &= 0.2008 \times 0.01655 \\ &= 3.323 \times 10^{-3} \text{ mol} \\ &= n(\text{C}_9\text{H}_8\text{O}_4) \text{ (1:1 ratio)} \\ m(\text{C}_9\text{H}_8\text{O}_4) &= n \times \text{Mm} \\ &= 3.323 \times 180.154 \text{ (} 9 \times 12.01 + 8 \times 1.008 + 4 \times 16.00 \text{)} \\ &= 0.5986 \text{ g} \\ &= 598.6 \text{ mg} \end{aligned}$$

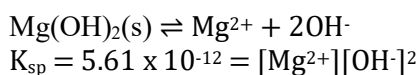
$$\begin{aligned} m(\text{C}_9\text{H}_8\text{O}_4) &= \frac{598.6}{2} \\ \text{Per tablet} &= 299.3 \text{ mg} \end{aligned}$$

The claim about the tablet is not true because the calculated value of 299.3mg is much greater than the stated claim of 150 mg

### Question 24 (4 marks)

- ( What is the maximum pH which can be obtained by dissolving  $\text{Mg}(\text{OH})_2$  in pure water?  
a ) 2

| Criteria                                                | Mark |
|---------------------------------------------------------|------|
| Correctly calculates $[\text{OH}^-]$ and the maximum pH | 2    |
| Correctly calculates $[\text{OH}^-]$                    | 1    |



$$\text{Let } x = [\text{Mg}^{2+}] = 2[\text{OH}^-]$$

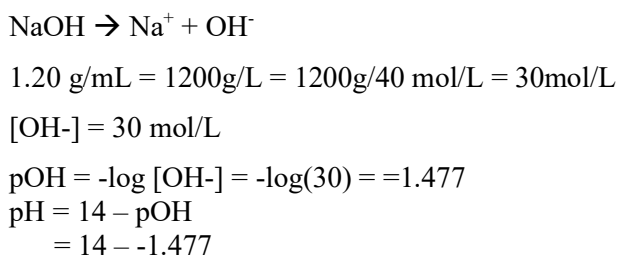
$$\begin{aligned} 5.61 \times 10^{-12} &= x(2x)^2 \\ 5.61 \times 10^{-12} &= 4x^3 \\ x &= \sqrt[3]{5.61 \times 10^{-12} / 4} = 0.000112 \end{aligned}$$

$$\begin{aligned} [\text{OH}^-] &= 2 \times 0.000112 \\ &= 0.000224 \text{ mol/L} \end{aligned}$$

$$\begin{aligned} \text{pOH} &= -\log [\text{OH}^-] = -\log(0.000224) = 3.65 \\ \text{pH} &= 14 - \text{pOH} \\ &= 14 - 3.65 \\ &= 10.34 \end{aligned}$$

- ( How does this compare with the maximum pH which can be obtained with NaOH?  
b (The solubility of NaOH is 1.20 g/mL at 25°C) ) 2

| Criteria                                                | Mark |
|---------------------------------------------------------|------|
| Correctly calculates $[\text{OH}^-]$ and the maximum pH | 2    |
| Correctly calculates $[\text{OH}^-]$                    | 1    |



= 15.5

**Question 25 (14 marks)**

( 

|                                                          |   |
|----------------------------------------------------------|---|
| $K = \frac{[\text{NO}_2]^2}{[\text{N}_2][\text{O}_2]^2}$ | 1 |
|----------------------------------------------------------|---|

 ) 1

a )

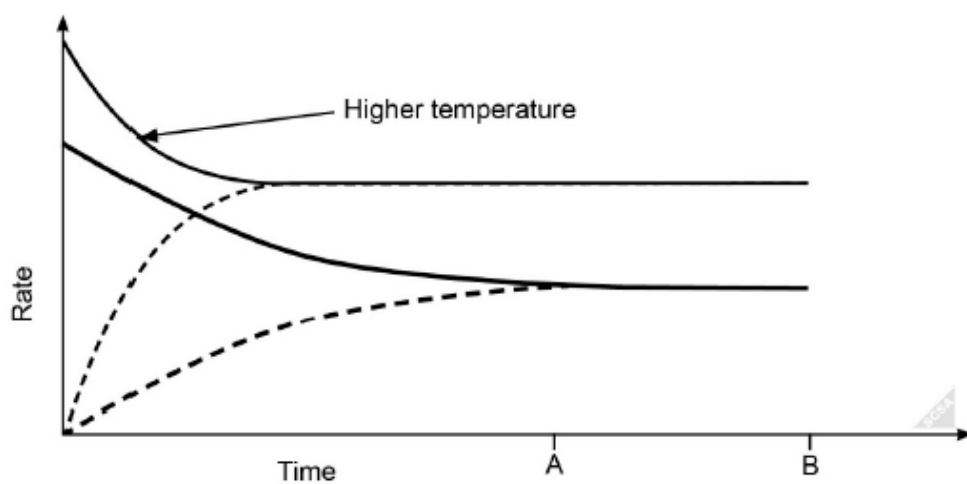
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b )

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i )

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( 2

i

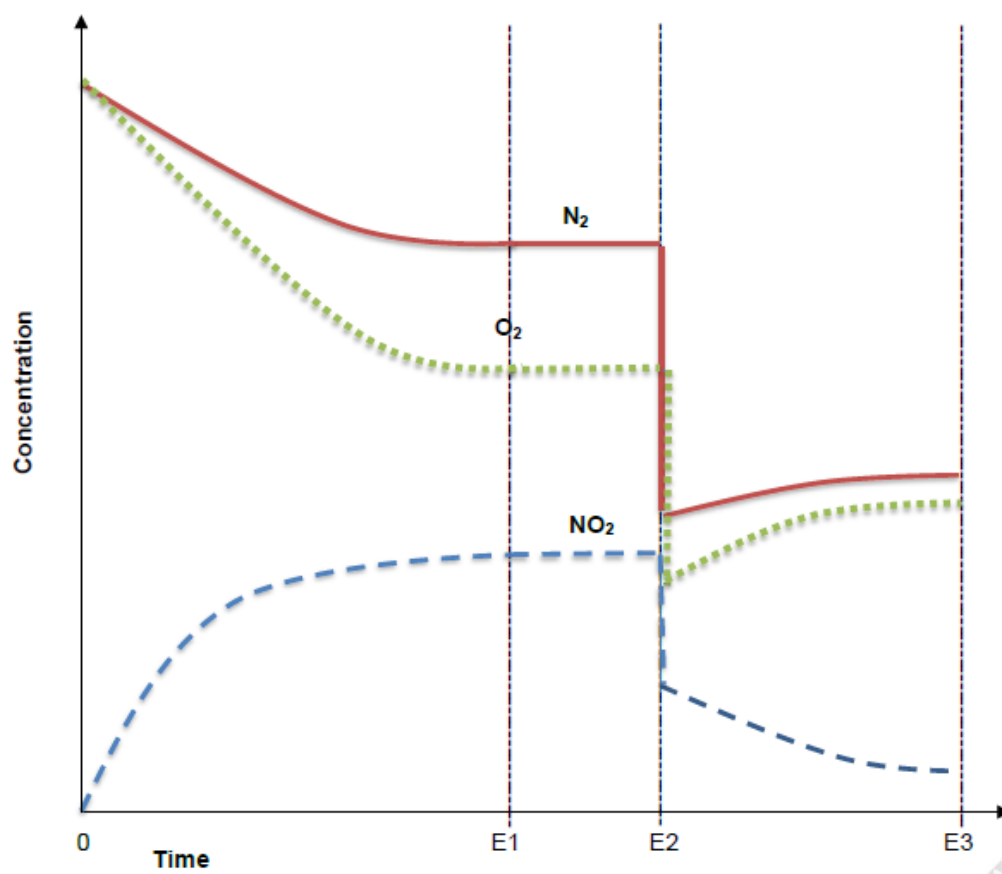
i

)

(

c Label clearly the line for each gas. 5

)



(  
d  
)

3

### Question 26 (5 marks)

- (a) What is the effect on pH when  $\text{OCl}^-$  ions are added to the water?

1

| Criteria                          | Mark |
|-----------------------------------|------|
| Correctly states the effect on pH | 1    |

The increase in the concentration of  $\text{OCl}^-$  causes the pH to increase.

- (b) At  $\text{pH} = 8.5$ , what is the ratio of  $[\text{OCl}^-] : [\text{HOCl}]$ ?

1

| Criteria                      | Mark |
|-------------------------------|------|
| Correct ratio at pH 8.5 given | 1    |

The ratio of  $[\text{OCl}^-] : [\text{HOCl}] = 9:1$  (from graph)

(c) Using your answer in part (b), calculate  $K_a$  for hypochlorous acid.

3

| Criteria                                                                                                                                                               | Mark |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Calculates $[\text{H}^+]$ using pH<br>AND Calculates $[\text{HOCl}]$ using ratio from part (b)<br>AND Correctly calculates $K_a$ with all working shown                | 3    |
| Calculates $[\text{H}^+]$ using pH AND Correct $K_a$ expression<br>OR<br>Correct $K_a$ expression AND Calculates correct $K_a$ value with incomplete working out shown | 2    |
| Correct $K_a$ expression OR Correctly calculate $[\text{H}^+]$<br>OR Calculates correct $K_a$ value with incomplete working                                            | 1    |

$$K_a = \frac{[\text{H}^+][\text{OCl}^-]}{[\text{HOCl}]}$$

$$[\text{H}^+] = 10^{-8.5} = 3.16 \times 10^{-9}$$

$$[\text{HOCl}] = \% \times [\text{H}^+]$$

$$= 10/90 \times 3.16 \times 10^{-9}$$

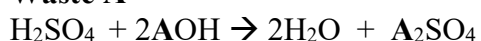
$$= 3.514 \times 10^{-10} \text{ molL}^{-1}$$

$$K_a = \frac{(3.16 \times 10^{-9})^2}{3.514 \times 10^{-10}} = 2.842 \times 10^{-8} = 2.8 \times 10^{-8} \text{ (2 s.f.)}$$

**Question 27 (5 marks)**

**5**

**Waste A**

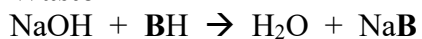


$$n(\text{H}_2\text{SO}_4) = cV = 0.233 \times 29.5/1000 = 0.0068735 \text{ mol}$$

$$n(\text{AOH}) = 2 \times n(\text{H}_2\text{SO}_4) = 2 \times 0.00687 = 0.013747 \text{ mol}$$

$$c(\text{AOH}) = n/v = 0.013747/0.025 = 0.54988 \text{ mol/L}$$

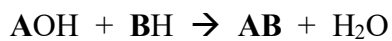
**Waste B**



$$n(\text{NaOH}) = cV = 0.274 \times 32.8/1000 = 0.0089872 \text{ mol}$$

$$n(\text{OH}^-) = 0.0089872 \text{ mol}$$

$$c(\text{BH}) = n/V = 0.0089872/0.025 = 0.359488 \text{ mol/L}$$



$$n(\text{BH})[\text{waste B in 100L}] = cv = 0.359488 \times 100 = 35.9488 \text{ mol}$$

$$n(\text{AOH}) = cv = 0.54988 \times v$$

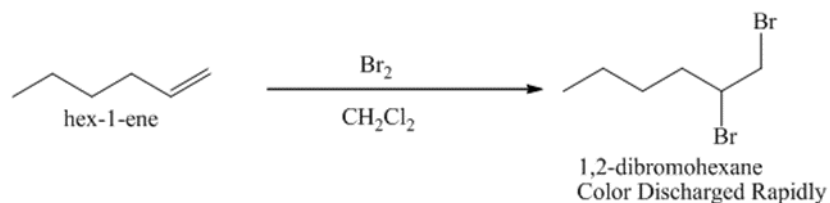
$$35.9488 \text{ mol} = 0.54988 \times v$$

$$V = 35.9488 \text{ mol} / 0.54988 \text{ mol/L} = 65.3757 \text{ L} = \mathbf{65.4 \text{ L}}$$

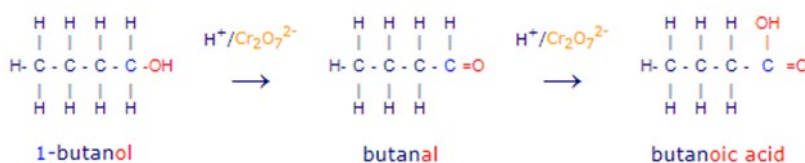
## Question 28 (4 marks)

4

Into three separate test-tubes transfer roughly 5 mL of each liquid, and add 5 drops of bromine water then shake. One liquid will turn the orange/brown bromine water colourless, this is 1-hexene. The other two liquids will not show a colour change. The reaction is as such:



For the remaining two liquids, transfer roughly 5mL to two fresh test-tubes and add 5 drops of acidified potassium dichromate solution then shake. The liquid that changes colour from orange to green is 1-butanol. The reaction is as such:

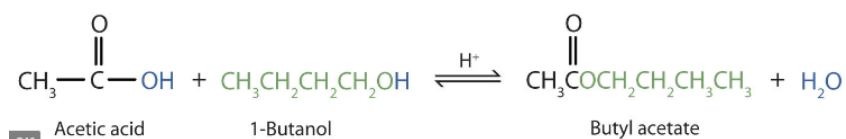
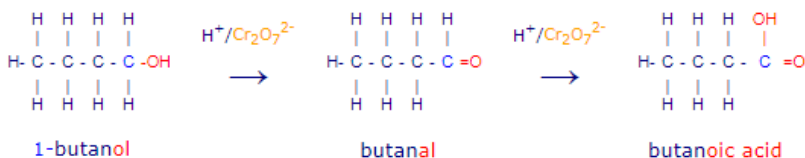


The other liquid will not change colour, which is 2-methyl-2-propanol.

### Other chemical tests possible:

#### 1-Butanol – primary alcohol

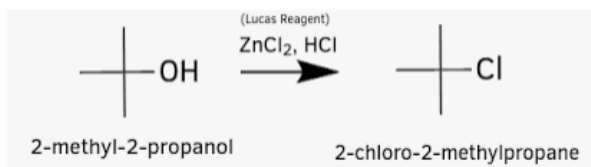
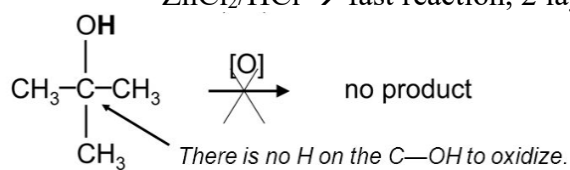
- React with -  $\text{MnO}_4^-/\text{H}^+$  → purple to colourless solution
- $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$  → orange solution changes to green
- a carboxylic acid to form a sweet smelling ester



#### 2-methyl – 2- propanol – tertiary alcohol

React with -  $\text{MnO}_4^-/\text{H}^+$  → purple solution remains unchanged, no reaction

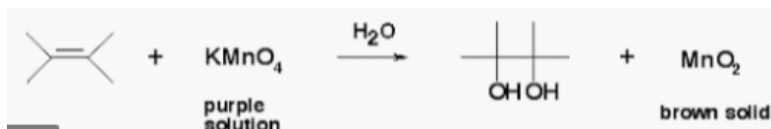
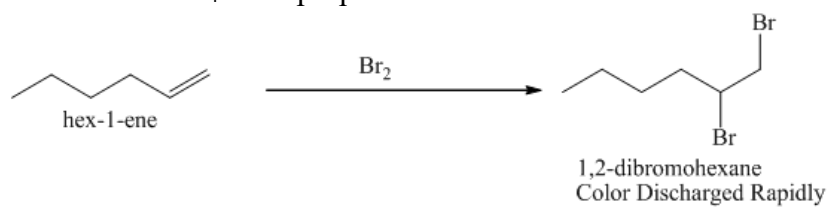
- $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+ \rightarrow$  orange solution remains unchanged, no reaction
- $\text{ZnCl}_2/\text{HCl} \rightarrow$  fast reaction, 2 layers form



### 1-hexene -unsaturated hydrocarbon

React with -  $\text{Br}_2$  or  $\text{BrOH}$  – orange/brown solutions turns colourless

- $\text{MnO}_4^-/\text{H}^+$  - purple colour turns brown



Question 29 (4 marks)

4

Kevlar is a strong nylon polymer consists of many strong intermolecular forces holding the polymer together thus it possesses the quality as strong and durable. This is due to the presence of hydrogen bonding, dipole-dipole & dispersion forces through the polymer. ✓

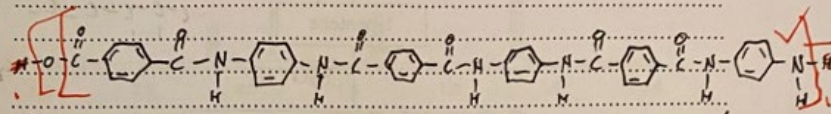
The strong electronegative nitrogen bonds with the strong electropositive hydrogen to form a very polar dipole moment across this bond. The type of permanent dipole-dipole moment can be further classified as hydrogen bonding and can attract other hydrogen bonds as shown below. (in red) ✓

Similarly the carbonyl ~~group~~ bond between C=O bond is also very polar due to its electronegativity difference allowing further dipole-dipole ~~moment~~ ~~and~~ attraction to occur.

Thus summarizing the presence of hydrogen bonding, dipole-dipole & dispersion force cause the intermolecular force of this polymer to be extremely strong and rigid allowing it to be used for ropes and fishing nets as it's <sup>very</sup> durable due to its chemical structure. ✓

Note: this answer doesn't have a segment of polymer

Kevlar is a condensation polymer, forming a polyamide with a strong peptide link, and multiple benzene crystals.



Evidence from the segment of Kevlar, Kevlar contains many benzene rings. This helps give Kevlar a strong crystalline structure, making it extremely durable, as a result of the various dispersion forces capable of being formed. **stacking**

Kevlar has electronegative oxygen from the peptide link. This will cause electron withdrawal and slightly negatively charge the other oxygen. As a result, Kevlar will be capable of forming dipole-dipole interactions with itself, along with the dispersion forces. Furthermore, Kevlar has electronegative nitrogen bonded to hydrogen. The internal electron withdrawal will cause nitrogen to become positive, and hydrogen to become negative. **Hydrogen Bond**

Figure 1.  $\begin{matrix} \delta^- N - H \delta^+ \\ \vdots \quad \vdots \\ \delta^+ H - N \delta^- \end{matrix}$  dipole attraction will allow Kevlar to form very strong hydrogen bonds, on top of the previous intermolecular forces. This makes Kevlar very strong and durable.

Furthermore, the benzene rings often create some branching, which gives Kevlar abrasion and impact resistance, as well as flexibility. On top of its durability and strength. As a result of proper bonds and structure, Kevlar is commonly used in military wear such as bullet proof suits or bullet proof vests.

Its ~~flexibility~~ flexibility allows it to be worn, but its strength and durability, as well as abrasion and impact resistance makes it suitable for military applications. As a result of ~~its~~ its structure and its ability to form various strong intermolecular forces, Kevlar has such strength and this explains its subsequent applications.

Markers notes:

Be careful when describing H-bonding. It's an intermolecular force between polymer chains not within the molecule.

Question was basically about the intermolecular forces present and the stacking ability of the chains that are fairly linear due to the aromatic ring.

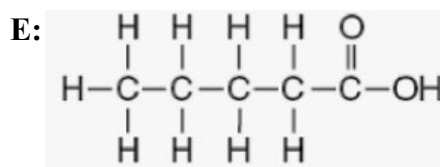
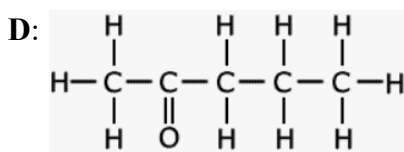
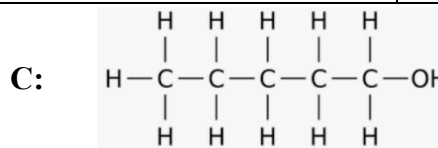
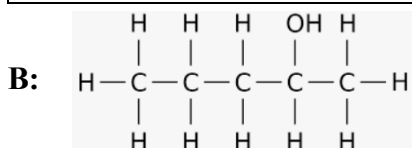
Linking structure to properties to uses.

### Question 30 (15 marks)

- (a) Draw the structural formulas of compounds B to F in the boxes provided.

5

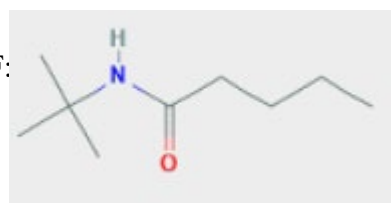
| Criteria                          | Mark |
|-----------------------------------|------|
| 1 mark for each correct structure | 5    |



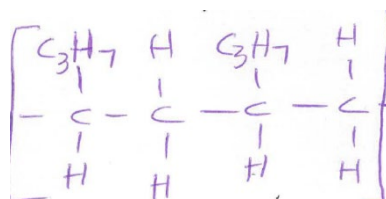
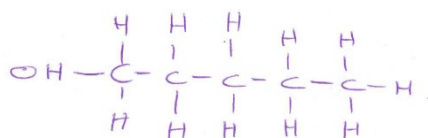
Aldehyde was accepted for E

If F was incorrect

F:

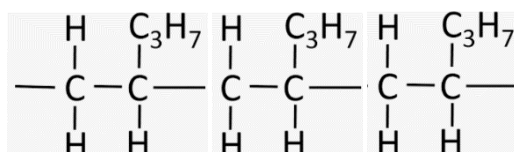


When drawing structures, ensure atoms have correct number of bonds to them. C-4, H-1, O-2. Don't do:



- (b) Draw the semi-structural formula, containing 3 monomers, of compound A in the box provided.

1



1 mark for correct polymer structure

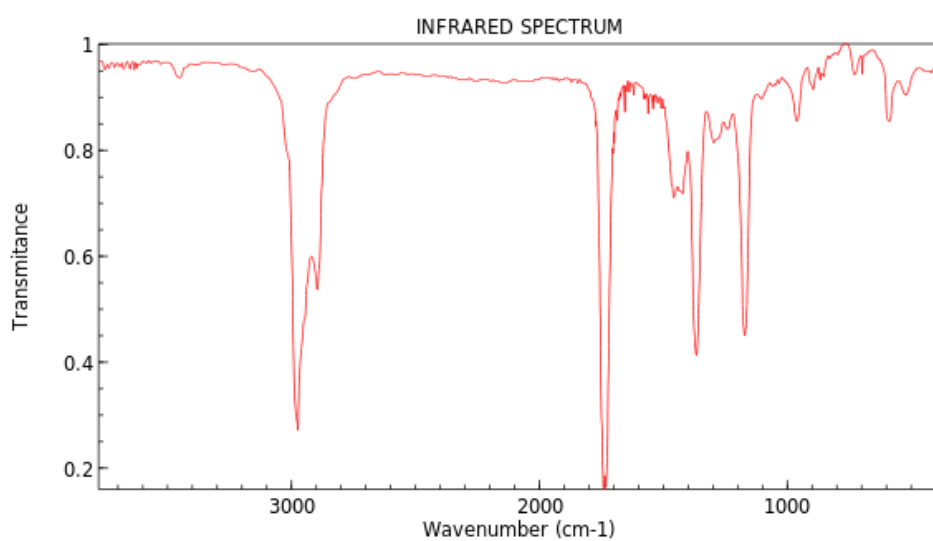
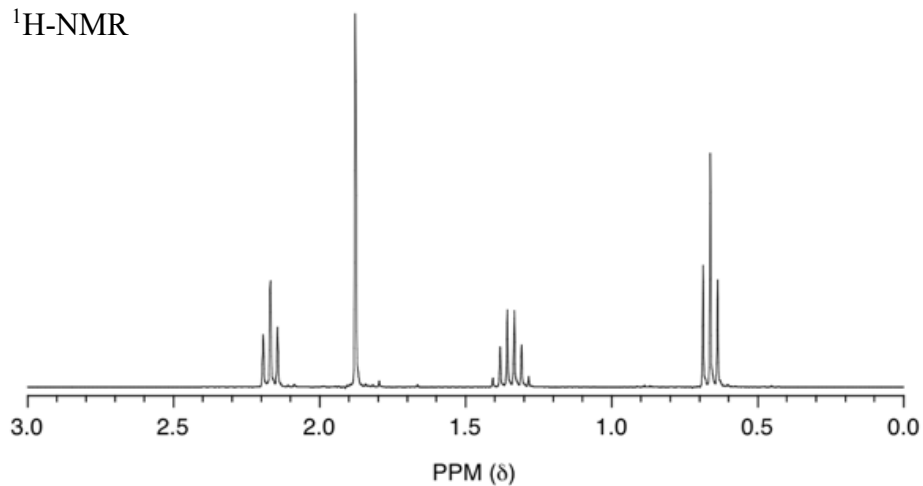
Done REALLY badly. Segment wasn't accepted if H's were missing. Laziness

Question 30 continues on the following page.

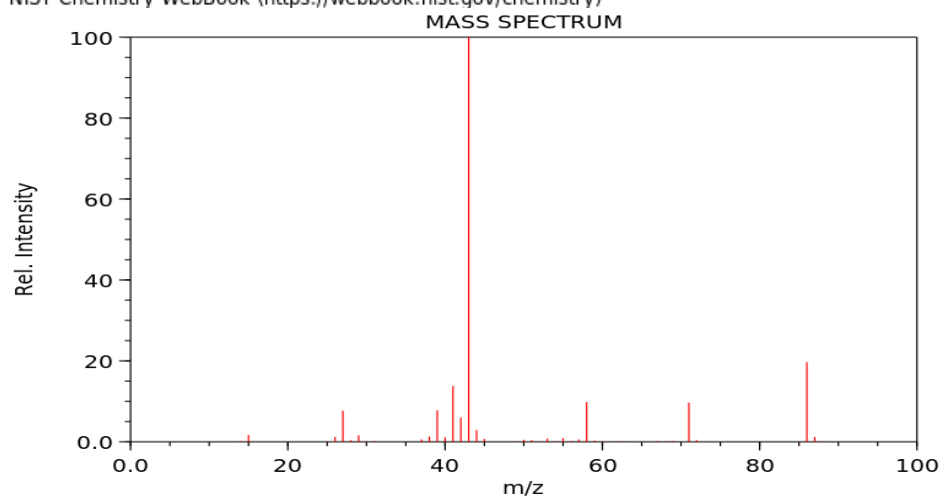
Question 30 continues.

- (c) To distinguish between compounds D and E, the student used spectroscopic analysis to determine the structure of the compound produced. He was provided with the following spectra.

$^1\text{H-NMR}$



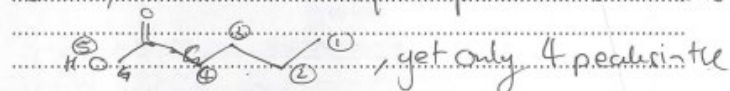
NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)



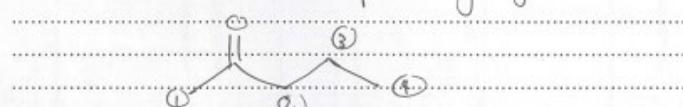
NIST Chemistry WebBook (<https://webbook.nist.gov/chemistry>)

Identify which compound from the previous synthetic pathway corresponds to the provided spectra. Justify your choice using the information from each spectrum. 6

Compound D. In the IR spectra, there is no acidic  $-OH$  broad peak at about  $2500-3000\text{ cm}^{-1}$  which would be expected if the compound was E, due to the carbonyl group. ~~The presence of~~ However, it is expected to be absent for D since there are no hydroxyl groups on D (ketone). The presence of  $C-H$  <sup>sharp</sup> peaks at about  $3000\text{ cm}^{-1}$  ( $2850-3000\text{ cm}^{-1}$ ) and a  $C=O$  sharp peak at about  $1740\text{ cm}^{-1}$  &  $(1650-1750\text{ cm}^{-1})$  is consistent with D's  $C=O$  group and  $C-H$  bonds. In E, there are 5 unique proton environments:



$^1H$  NMR (5 is expected instead). However, D does have the consistent 4 unique hydrogen environments:



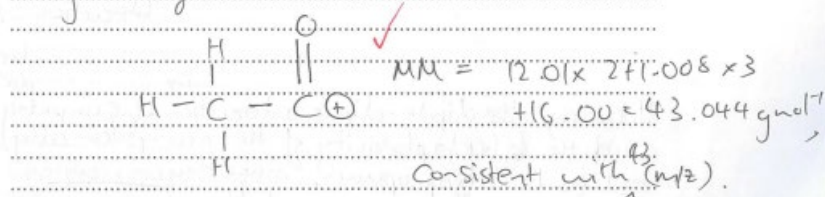
Additionally, the splitting patterns are consistent with D as labelled in the diagram:

- 3H on ① are split not split (singlet), adjacent no H atoms being  $<30^\circ$  bonds away
- 2H on ② split into triplet by 3H on ①
- 2H on ③ split into sextet by 2H on ② and 3H on ④
- 3H on ④ split into triplet by 2H on ③

The chemical shifts are also inconsistent with E.

A  $9.0-13.0\text{ ppm}$  would be expected for the acidic proton, yet the highest is about  $2.18\text{ ppm}$ .

The mass spectrum is consistent with D. The molecular ion peak at about  $86\text{ (m/z)}$  is consistent with molar mass of D,  $MM = 12.01 \times 5 + 1.008 \times 10 + 16 = 86.13\text{ g mol}^{-1}$ . The base peak can be provided by the possible fragment of:



E cannot provide a fragment accounting for this  $43\text{ (m/z)}$  base peak. Hence, it is D.

This question was answered really well.

IR - A lot of answers told me what wasn't in the spectra rather than what was in the spectra ie left out analysing the peak that was at  $2900\text{ cm}^{-1}$

MS – must refer to both the parent ion (not mother), the base peak and the fragment responsible for it. Both Compounds D and E contain the propyl fragment.

NMR – Needed to assign peaks based on splitting pattern and the chemical shift and state why they were shifted downfield.

Terminology – work on this ie triplet – not 3 splits (this would cause 4 peaks), singlet, triplets and sextet.

The IUPAC name for Compound D is: Pentan-2-one or 2-pentanone

**1 mark for correct name**

Pentan-2-one or 2-pentanone

Note: pent-2-one incorrect

In a  $^{13}\text{C}$  NMR spectrum of Compound D, how many signals would you expect? Why?

***Sample Answer:***

5 signals as there are five different environments due to the presence of the carbonyl carbon and no symmetry is present.

(d)

1

(e)

2

### Question 31 (5 marks)

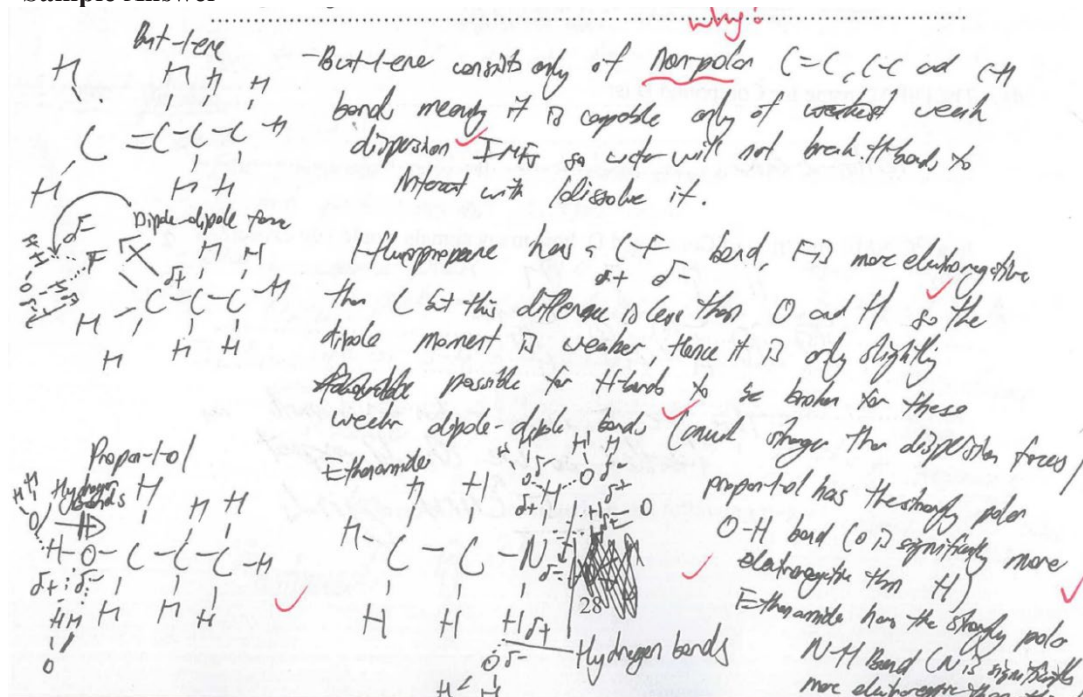
Some properties of but-1-ene, 1-fluoropropane, propan-1-ol and ethanamide are shown in the table.

| Compound        | Molar mass ( $\text{g mol}^{-1}$ ) | Solubility in water |
|-----------------|------------------------------------|---------------------|
| but-1-ene       | 56                                 | insoluble           |
| 1-fluoropropane | 62                                 | slightly soluble    |
| propan-1-ol     | 60                                 | soluble             |
| ethanamide      | 59                                 | soluble             |

Explain the different water solubilities of these compounds. Support your answer with at least ONE labelled diagram.

5

#### Sample Answer



Q) 31: Because both propan-1-ol and ethanamide have these strong dipole moments, they can also engage in hydrogen bonding with water. Thus, it would be thermodynamically and entropically favourable for water to break its H-bonds to form new H-bonds with both propan-1-ol and ethanamide and thus dissolve them.

*Ethanamide also has multiple H-bonding sites*



**Question 32 (11 marks)**

- (a) Calculate the experimental molar enthalpy of combustion ( $\Delta H_c$ ) of propanol when 0.543 g propanol was used to raise the water temperature from 19.0°C to 37.5°C. 4  
 $MM(C_3H_7OH) = 60.10 \text{ g mol}^{-1}$

$$\text{Moles}(\text{propanol}) = 0.543/60.11 = 0.009033 \text{ mols}$$

$$\begin{aligned} q &= mC\Delta T \\ &= 112 \times 4.18 \times 18.5 \\ &= 8660.96 \text{ J} \end{aligned}$$

$$\begin{aligned} \Delta H_c &= -q/n = -8660.96/0.009033 \\ &= -958767 \text{ J/mol} \\ &= -959 \text{ kJ/mol (3 s.f.)} \end{aligned}$$

| Criteria                                                                                                                                                                               | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Provides a change that would improve accuracy</li> <li>Gives a reason for the low molar enthalpy of combustion related to the change</li> </ul> | 2     |
| <ul style="list-style-type: none"> <li>Provides some relevant information</li> </ul>                                                                                                   | 1     |

*Sample answer:*

The experimental value for the molar enthalpy of combustion is low, primarily due to heat loss to the environment.

- (b) A change that would reduce heat loss and improve accuracy is moving the spirit burner closer to the beaker. 2  
 (There are many other possible improvements that are acceptable.)

The molar heat of combustion of propan-1-ol,  $C_3H_7OH$ , is  $2021 \text{ kJ mol}^{-1}$ .

Calculate the mass of  $C_3H_7OH$ , in tonnes, required to produce 9 605 MJ of energy.

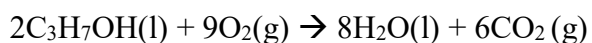
**Answer:**

$$\begin{aligned} \Delta H_c &= -q/n \\ 2021 \times 10^3 &= 9600 \times 10^6/n \\ n &= 9600 \times 10^6/2021 \times 10^3 \text{ moles} \\ &= 4750.124 \text{ mol} \\ m &= 4750.124 \times 60.10 = 285482.452\text{g} \\ &= 0.285 \text{ tonnes} \end{aligned}$$

- (c) One way of minimising the greenhouse effect is to minimise the amount of carbon dioxide formed per kilojoule of energy produced. 3

Using the theoretical  $\Delta H_c$ , calculate the amount of energy released, in kJ, per mole of carbon dioxide produced during the complete combustion of propan-1-ol.

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| Balanced chemical equation with states<br>Correct amount of energy calculated | 2     |
| Balanced chemical equation OR<br>Correct amount of energy calculated          | 1     |



Energy per mole of  $\text{CO}_2 = 2021/3 = 673.7$  kJ per mole of  $\text{CO}_2$  released

- (d) 2